MMDU MULLANA OFFICIAL SYLLABUS • BPHY-001 • UNIT 1

Unit 1: Wave Optics & Lasers (Complete Theory, Derivations & Quantum Systems)

Subject: Applied Physics (BPHY-001) • **Exam Weightage:** 26 - 30 Marks • **Study Time:** 40 mins

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1. Principles of Wave Optics & Newton's Rings (Complete Derivation)

1. Huygens' Principle & Superposition of Waves

Huygens' Principle forms the physical basis of wave optics:

1. Every point on an advancing primary wavefront serves as an independent secondary source of spherical wavelets radiating forward with wave speed c.

2. The envelope surface tangential to these forward wavelets at any later time $t + \Delta t$ establishes the new primary wavefront.

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2. Division of Wavefront vs. Division of Amplitude

Dimension	Division of Wavefront	Division of Amplitude
Physical Mechanism	Wavefront from a localized source is physically divided into two by apertures/prisms.	Wave amplitude is split by partial reflection and partial refraction at boundaries.
Source Geometry	Demands a narrow pinhole or slit source.	Can utilize broad, extended light sources.
Interference Fringes	Fringes are non-localized in space.	Fringes are localized (e.g. localized at film surface).
Prototypical Systems	**Young's Double Slit (YDSE), Fresnel Biprism, Lloyd's Mirror.**	**Newton's Rings, Thin Dielectric Films, Michelson Interferometer.**

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3. Newton's Rings: Derivation of Ring Diameters

When a plano-convex lens of large radius of curvature R rests upon an optically flat glass plate, an air film of circular symmetry and varying thickness t is formed.

> ✅ **FUNDAMENTAL LAW / RULE:**
> **Stokes' Phase Shift Theorem:**

> When light reflects at the boundary of an optically denser medium (air-to-glass interface at the flat plate), an abrupt phase reversal of π radians occurs, introducing an additional optical path difference of $\pm\lambda/2$.

For normal incidence ($\cos r = 1$) in an air film ($\mu = 1$), the net optical path difference is:

$$\Delta = 2t - \frac{\lambda}{2}$$

From the geometry of a sphere of radius R enclosing contact radius r_n and film thickness t:

$$r_n^2 = t(2R - t) \approx 2Rt \implies 2t = \frac{r_n^2}{R} = \frac{D_n^2}{4R}$$

where $D_n = 2r_n$ is the diameter of the nth fringe ring.

(A) Condition for Dark Rings (Destructive Interference)

$$\Delta = 2t - \frac{\lambda}{2} = (2n-1)\frac{\lambda}{2} \implies 2t = n\lambda$$

$$\frac{D_n^2}{4R} = n\lambda \implies D_n = 2\sqrt{n\lambda R} \implies D_n \propto \sqrt{n}$$

Conclusion: Diameters of dark rings are directly proportional to square roots of consecutive natural numbers ($0, 1, 2, 3\ldots$).

(B) Condition for Bright Rings (Constructive Interference)

$$\Delta = 2t - \frac{\lambda}{2} = n\lambda \implies 2t = (2n-1)\frac{\lambda}{2}$$

$$\frac{D_n^2}{4R} = (2n-1)\frac{\lambda}{2} \implies D_n = \sqrt{2(2n-1)\lambda R} \implies D_n \propto \sqrt{2n-1}$$

Conclusion: Diameters of bright rings are proportional to square roots of consecutive odd numbers ($1, 3, 5\ldots$).

(C) Measurement of Wavelength λ & Refractive Index μ

$$\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR} \qquad \mu = \frac{(D_{n+p}^2 - D_n^2)_{\text{air}}}{(D_{n+p}^2 - D_n^2)_{\text{liquid}}}$$

Why is the center dark in reflection? At point of contact O, film thickness $t = 0$. Hence path difference $\Delta = -\lambda/2$, fulfilling the exact condition for destructive interference!

2. Diffraction: Fresnel vs. Fraunhofer & Single Slit Phasor Derivation

1. Fresnel vs. Fraunhofer Diffraction (12-Point Analysis)

#	Parameter	Fresnel Diffraction	Fraunhofer Diffraction
1	**Source Distance**	Source is at **finite distance** from diffracting aperture.	Source is effectively at **infinite distance**.
2	**Screen Distance**	Screen is at **finite distance**.	Screen is effectively at **infinite distance**.
3	**Incident Wavefront**	Spherical or cylindrical wavefront.	Strictly **plane wavefront**.
4	**Diffracted Wavefront**	Spherical or cylindrical.	Strictly **plane wavefront**.
5	**Optical Lenses**	No lenses required.	Requires two convex lenses (collimator & focus).
6	**Phase Variations**	Quadratic phase terms (Fresnel integrals).	Linear phase variations across aperture.
7	**Mathematical Rigor**	Fresnel zones / Cornu spiral integration.	Fourier transform / simple phasor summation.
8	**Fringe Geometry**	Shadows of aperture; non-uniform spacing.	Spectral patterns; regular minima & maxima.
9	**Center of Pattern**	Can be bright or dark based on zone count.	Always a bright **Central Principal Maximum**.
10	**Real-Life Occurrence**	Edge of razor blade, small circular obstacle.	Spectrometers, telescope apertures, diffraction gratings.
11	**Observational Ease**	Requires critical alignment, no optical bench.	Standard spectrometer telescope configuration.
12	**Energy Concentration**	Dispersed over irregular shadow fringes.	Over **95%** of energy focused in central maximum.

2. Fraunhofer Single Slit Diffraction Derivation

A slit of width a is illuminated normally by a plane wave of wavelength λ. Dividing the slit into N equal elements of width $dx = a/N$, the resultant amplitude R at angle θ obtained by phasor addition is:

$$R = A_0 \frac{\sin \alpha}{\alpha} \quad \text{where } \alpha = \frac{\pi a \sin \theta}{\lambda}$$

The intensity distribution is:

$$I(\theta) = I_0 \left(\frac{\sin \alpha}{\alpha}\right)^2$$

- **Central Maximum ($\theta = 0$):** As $\alpha \to 0$, $\lim_{\alpha \to 0} \frac{\sin \alpha}{\alpha} = 1 \implies I = I_0$.
- **Minima Condition:** $\sin \alpha = 0$ while $\alpha \neq 0 \implies \alpha = \pm m\pi$ ($m = 1, 2, 3...$):

$$a \sin \theta = \pm m\lambda$$

- **Secondary Maxima Condition:** $\frac{dI}{d\alpha} = 0 \implies \tan \alpha = \alpha$. Solved graphically: $\alpha \approx \pm 1.43\pi, \pm 2.46\pi, \pm 3.47\pi$.
- **Intensity Ratio:** $I_0 : I_1 : I_2 : I_3 = 1 : 0.045 : 0.016 : 0.008$.
- **Central Angular Width:** $2\theta = \frac{2\lambda}{a}$ (Linear width on screen at focal length f is $2f\lambda/a$).

3. Polarization of Light & Birefringence

1. Polarization Fundamentals

In electromagnetic waves, the electric field $\vec{E}$ vibrates transversely to the propagation axis.

- **Unpolarized Light:** Symmetrical vibrations in all planes perpendicular to the propagation ray.
- **Linearly (Plane) Polarized Light:** Electric field vibrations restricted strictly to a single plane.

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2. Brewster's Law & Polarizing Angle

When unpolarized light is incident on a transparent dielectric medium at the polarizing angle i_p, the reflected ray is 100% plane-polarized perpendicular to the plane of incidence.

- Reflected ray and refracted ray are mutually perpendicular: $i_p + r = 90° \implies r = 90° - i_p$.
- From Snell's Law: $\mu = \frac{\sin i_p}{\sin r} = \frac{\sin i_p}{\sin(90° - i_p)} = \frac{\sin i_p}{\cos i_p} = \tan i_p$.

$$\mu = \tan(i_p)$$

3. Malus's Law

When plane-polarized light of intensity I_0 passes through an analyzer oriented at angle θ relative to the transmission axis of the polarizer:

$$I = I_0 \cos^2 \theta$$

- When $\theta = 0°$ (parallel axes): $I = I_0$ (maximum transmission).
- When $\theta = 90°$ (crossed nicols): $I = 0$ (complete extinction).

4. Birefringence & Retardation Plates

In anisotropic crystals (e.g. Calcite $CaCO_3$, Quartz), an incident ray splits into an **Ordinary Ray (O-ray)** obeying Snell's Law and an **Extraordinary Ray (E-ray)** with direction-dependent velocity.

- **Quarter-Wave Plate (QWP):** Introduces path difference $\lambda/4$ (phase shift $\pi/2$):

$$t = \frac{\lambda}{4(\mu_o - \mu_e)}$$

- **Half-Wave Plate (HWP):** Introduces path difference $\lambda/2$ (phase shift π):

$$t = \frac{\lambda}{2(\mu_o - \mu_e)}$$

4. Quantum Laser Physics: Einstein Coefficients & He-Ne Systems

1. Fundamental Characteristics of Laser Light

LASER stands for **Light Amplification by Stimulated Emission of Radiation**.

1. **Extreme Monochromaticity:** Spectral linewidth $\Delta\lambda \approx 10^{-6}$ Å.

2. **Directionality & Low Divergence:** Beam divergence angle $\theta < 1$ mrad.

3. **High Coherence:** Exceptional temporal coherence length (kilometers) and spatial wavefront coherence.

4. **Enormous Intensity:** Focused laser beams exceed solar core flux density ($> 10^{12}$ W/cm^2).

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2. Einstein's A and B Coefficients Derivation

Consider two atomic levels E_1 and E_2 in thermal equilibrium with radiation density $u(\nu)$ at temperature T:

1. **Absorption:** $R_{12} = B_{12} N_1 u(\nu)$

2. **Spontaneous Emission:** $R_{21\,\text{sp}} = A_{21} N_2$

3. **Stimulated Emission:** $R_{21\,\text{st}} = B_{21} N_2 u(\nu)$

At thermodynamic equilibrium: Rate of Upward Transitions = Rate of Downward Transitions:

$$B_{12} N_1 u(\nu) = A_{21} N_2 + B_{21} N_2 u(\nu) \implies u(\nu) = \frac{A_{21}}{B_{12}\frac{N_1}{N_2} - B_{21}}$$

From Maxwell-Boltzmann statistics: $N_1/N_2 = e^{(E_2 - E_1)/k_B T} = e^{h\nu/k_B T}$:

$$u(\nu) = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{B_{12}}{B_{21}} e^{h\nu/k_B T} - 1}\right]$$

Comparing directly with **Planck's Law of Blackbody Radiation**:

$$u(\nu) = \frac{8\pi h \nu^3}{c^3} \left[\frac{1}{e^{h\nu/k_B T} - 1}\right]$$

We arrive at two fundamental Einstein identities:

$$B_{12} = B_{21} \qquad \frac{A_{21}}{B_{21}} = \frac{8\pi h \nu^3}{c^3} \propto \nu^3$$

Physical Implication: The probability of spontaneous emission increases dramatically with frequency ν^3, explaining why

gamma-ray and X-ray lasers require astronomical pumping powers compared to microwave MASERs.

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3. Population Inversion & Metastable States

Under thermal equilibrium, $N_1 > N_2$. For net optical gain, one must enforce **Population Inversion ($N_2 > N_1$)**. This is achieved by pumping atoms into a **Metastable State** having an unusually long lifetime ($\sim 10^{-3}$ s compared to normal 10^{-8} s).

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4. Four-Level Helium-Neon (He-Ne) Gas Laser

- **Active Medium:** Gas mixture of Helium and Neon in a 10:1 ratio at low pressure (~ 1 torr).
- **Pumping Mechanism:** High-voltage DC electrical discharge. Fast electrons excite He atoms from ground state 1^1S_0 to metastable levels 2^1S_0 (20.61 eV) and 2^3S_1 (19.82 eV).
- **Resonant Energy Transfer:** Excited He atoms collide inelastically with ground state Neon atoms, transferring energy with minimal mismatch ($\Delta E \approx 0.05$ eV) to excite Neon into $3s$ (20.66 eV) and $2s$ (19.78 eV) states.
- **Laser Emission:** Stimulated radiative transition occurs from Neon $3s \rightarrow 2p$, emitting a continuous, highly coherent red laser beam at:

> $$\lambda = 632.8 \text{ nm}$$

- **Depopulation:** Fast spontaneous decay occurs from $2p \rightarrow 1s$, followed by non-radiative de-excitation to the ground state via physical collisions with the capillary tube walls.